



LIONS CLUB OF BARODA RISING STAR

Club No. 179323 | District 3232 F1 | Region 6 | Zone 1 | Vadodara, India

System Workflow Blueprint

CRM · Mobile App · Website — every integration & configuration, wired together

PUBLIC WEBSITE

Marketing, donations,
invoices & member portal

ADMIN / CRM

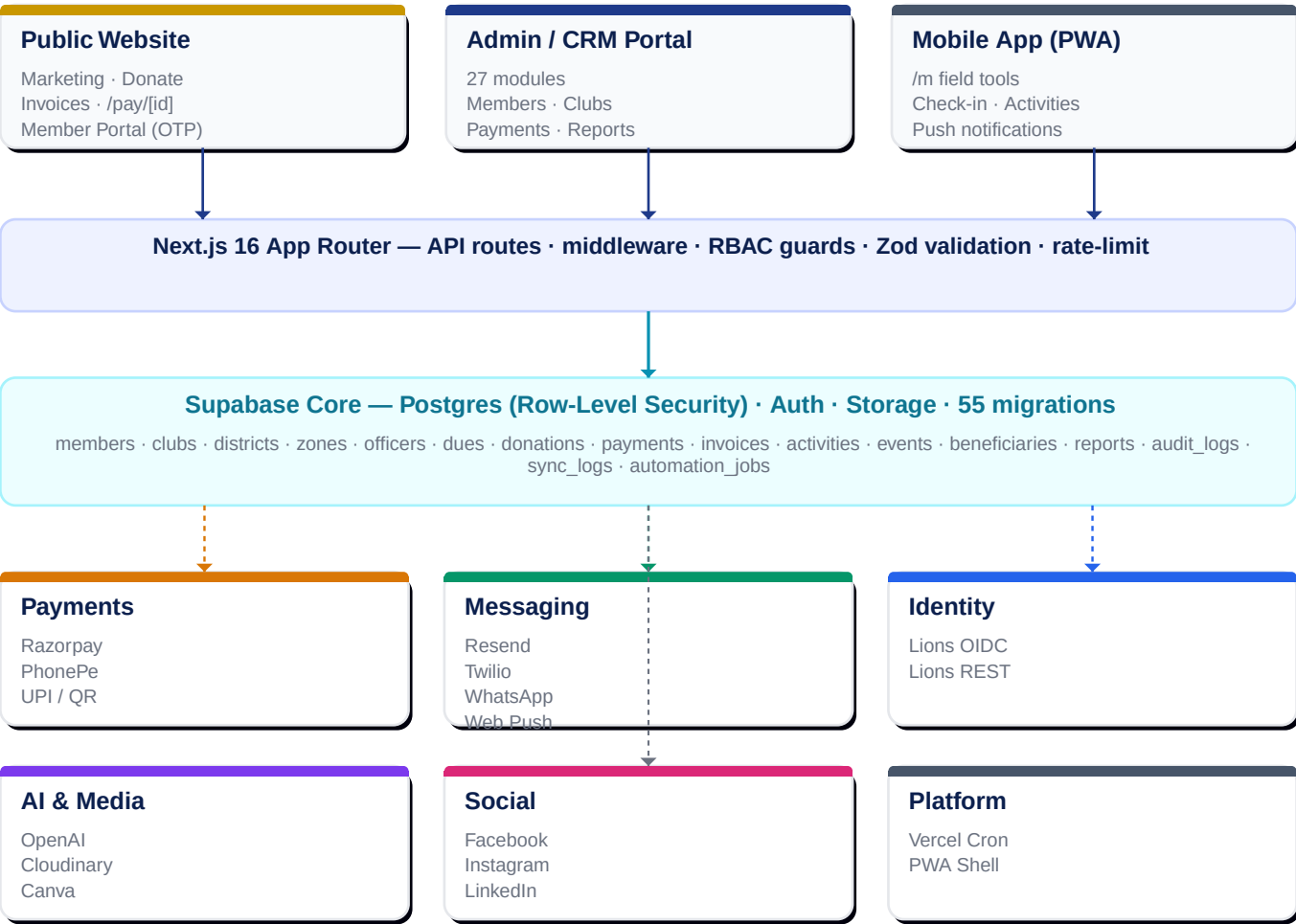
27 modules: members, clubs,
districts, payments, sync

MOBILE APP (PWA)

Field tools: check-in, activity
logging, beneficiaries, push

System Architecture — wired overview

Three user surfaces converge on one Next.js API layer and a single Supabase core, which fans out to every external integration.



How to read this
Solid arrows = synchronous request path (browser -> API -> database). Dashed arrows = the database/automation layer reaching out to external providers (webhooks, cron-driven jobs, AI calls). Every integration degrades gracefully when unconfigured — the feature it powers simply hides.

The three surfaces

One codebase, one database — served as a website, an admin CRM, and a mobile PWA.

1 · Public Website

Server-rendered, SEO-optimised marketing site and supporter-facing flows. Routes under (public).

- Marketing: Home, About, Activities, Events, Impact, Campaigns, Media, Stories, Blog.
- Giving: /donate (Razorpay + UPI), /pay/[id] one-tap invoice payment with dynamic UPI QR.
- Invoices: /invoices/lookup self-service receipt & invoice lookup.
- Member Portal: /portal OTP login, communication preferences, logout.
- Trust: /privacy, /terms, contact form (rate-limited, stored to DB).

2 · Admin / CRM Portal

Auth-gated control plane under /admin (27 modules). RBAC-scoped; every privileged action is audited.



3 · Mobile App — PWA

Installable progressive web app at /m with offline shell, home-screen icon and push. Built for Lions in the field.

- Home dashboard + bottom tab bar; service worker caches the app shell.
- QR check-in scanner for event attendance; activity logging with photo upload + AI narrative.
- Beneficiary intake, member & club directory, reports, AI greetings composer, networking.
- Push toggle (VAPID) for broadcast / per-member notifications.
- App shortcuts: Log activity · QR check-in · Add beneficiary · Reports.

Integration & configuration registry

Every external service the platform talks to (20 total), the environment keys that wire it, and what breaks when it is missing.

Identity & Access

4 services

Supabase Auth

Email sign-in for admins & members; backs /login and the member portal.

KEYS: `NEXT_PUBLIC_SUPABASE_URL` `NEXT_PUBLIC_SUPABASE_ANON_KEY`

Supabase Service Role

Server-only admin client for cron, sync, reports & webhooks (bypasses RLS).

KEYS: `SUPABASE_SERVICE_ROLE_KEY`

Lions International SSO (OIDC)

PKCE + JWKS + discovery, role mapping, MyLCI claims. Configure in-app or via env.

KEYS: `LIONS_OIDC_ISSUER` `LIONS_OIDC_CLIENT_ID` `LIONS_OIDC_REDIRECT_URI` `LIONS_OIDC_CLIENT_SECRET` `LIONS_OIDC_SCOPES`

Lions International REST API

MyLCI-shape adapter syncing districts, clubs & members. Dry-run when unset.

KEYS: `LIONS_API_BASE_URL` `LIONS_API_KEY` `LIONS_API_ACCESS_TOKEN` `LIONS_API_DISTRICT_CODE`

Database

1 service

Supabase Postgres + Storage

Primary datastore with Row-Level-Security on every table; media buckets.

KEYS: `SUPABASE_PROJECT_ID` `NEXT_PUBLIC_SUPABASE_URL`

Payments

3 services

Razorpay

Card / netbanking / UPI hosted checkout, refunds, recurring invoices, webhook reconciliation.

KEYS: `RAZORPAY_KEY_ID` `RAZORPAY_SECRET` `RAZORPAY_WEBHOOK_SECRET` `NEXT_PUBLIC_RAZORPAY_KEY_ID`

PhonePe

PhonePe Standard Checkout for UPI-first flows with webhook auto-verification.

KEYS: `PHONEPE_MERCHANT_ID` `PHONEPE_SALT_KEY` `PHONEPE_SALT_INDEX`

UPI Deep-links + QR

PhonePe / GPay / Paytm intent URLs and dynamic UPI QR on /pay/[id].

KEYS: `UPI_VPA` `UPI_PAYEE_NAME` `NEXT_PUBLIC_UPI_VPA`

Messaging

4 services

Resend (Email)

Transactional email — receipts, payment confirmations, OTP, portal links.

KEYS: `RESEND_API_KEY` `RESEND_FROM_EMAIL`

Twilio (SMS / WhatsApp)

SMS and WhatsApp Business via Twilio sandbox / approved sender.

KEYS: `TWILIO_ACCOUNT_SID` `TWILIO_AUTH_TOKEN` `TWILIO_WHATSAPP_FROM`

WhatsApp Business Cloud

Meta WhatsApp Cloud API — preferred for outbound notifications & templates.

KEYS: WHATSAPP_BUSINESS_TOKEN WHATSAPP_BUSINESS_PHONE_ID

Web Push (VAPID)

PWA push — broadcast / topic / per-member. Keypair auto-generated on install.

KEYS: VAPID_PUBLIC_KEY VAPID_PRIVATE_KEY VAPID_SUBJECT

Artificial Intelligence

1 service

OpenAI (Chat + Vision)

Narrative writer (EN+GU), greetings, creative builder, club insights, dedupe, UPI/expense OCR.

KEYS: OPENAI_API_KEY OPENAI_MODEL

Social

3 services

Meta — Facebook Pages

Auto-post to the club Facebook Page with images & captions.

KEYS: META_ACCESS_TOKEN FACEBOOK_PAGE_ID META_APP_ID

Meta — Instagram Business

Auto-post to the club Instagram Business account.

KEYS: META_ACCESS_TOKEN INSTAGRAM_BUSINESS_ID

LinkedIn Organization

Auto-post to the club LinkedIn organization page.

KEYS: LINKEDIN_ACCESS_TOKEN LINKEDIN_ORGANIZATION_URN LINKEDIN_CLIENT_ID

Media

2 services

Canva Connect

Canva Connect API for branded creatives / event posters.

KEYS: CANVA_CLIENT_ID CANVA_CLIENT_SECRET CANVA_API_KEY

Cloudinary

Hosted media CDN for activity photos, before/after galleries, event covers.

KEYS: CLOUDINARY_CLOUD_NAME CLOUDINARY_API_KEY CLOUDINARY_API_SECRET

Platform

2 services

Vercel Cron

Scheduled automation + monthly/quarterly/half-yearly/yearly reports. Secret auto-provisioned.

KEYS: CRON_SECRET

PWA Shell

Mobile app at /m — installable icon, offline shell cache, push notifications.

KEYS: (baked into build)

Required keys are shown in red, optional in grey (suffix removed). Identity / Lions services and OpenAI, Web Push and Cron can also be configured in-app — their secrets are stored encrypted (AES-256-GCM) in the database and take precedence rules described under Security.

Key data flows

The wired sequences that connect surfaces, the API, the database and external providers.

Donation / invoice payment

- 1 Browser POSTs /api/donations/intent (or /pay/[id]) -> creates donation + payment rows, returns a Razorpay order (or renders a UPI QR / PhonePe checkout).
- 2 Supporter pays in the hosted checkout or their UPI app.
- 3 POST /api/donations/verify checks the signature, marks the payment captured, sets dues/invoice paid, enqueues a receipt job.
- 4 Webhook /api/webhooks/razorpay (HMAC-verified) and /api/webhooks/phonepe back up the client flow for captured / failed events.
- 5 Automation engine renders the 80G-style PDF receipt and emails (Resend) + WhatsApps it to the donor.

Lions SSO login -> member upsert

- 1 /api/auth/oidc/login generates PKCE + state + nonce, redirects to the Lions IdP.
- 2 IdP returns code to /api/auth/oidc/callback -> verifies state, exchanges code for tokens (PKCE), verifies the ID token via JWKS.
- 3 fetchUserInfo() -> upsert oauth_accounts (linked by lions_member_id then email), promotes role claims onto members.
- 4 createSession() sets a signed session cookie; writeAudit("oauth.login"); redirect -> /admin.

Sync engine (CSV · REST · webhook)

- 1 Officer (RBAC.sync.trigger) uploads a CSV or triggers a REST pull; or the Lions provider POSTs a signed payload to /api/webhooks/lions.
- 2 runSyncJob() opens a sync_logs row (running); the entity adapter validates each row with Zod.
- 3 Valid rows upsert members / clubs / districts; failures are captured with reasons; a paginated cursor is persisted for REST.
- 4 sync_logs is finalised with counts; writeAudit(sync.success | partial | failed). The sync-worker cron drains the queue daily.

Automation, cron & notifications

- 1 Events (new member, captured donation, officer appointment, dues due) enqueue rows in automation_jobs.
- 2 Vercel Cron hits /api/cron/automation (Bearer CRON_SECRET) on schedule; the engine claims and runs handlers.
- 3 Handlers send email (Resend), WhatsApp (Twilio / Meta Cloud), SMS, and Web Push; each writes a comms log.
- 4 Report crons generate monthly / quarterly / half-yearly / yearly PDFs; action-item, donor-pack and sync-worker crons follow.

AI & OCR assist

- 1 Activity narrative (EN+GU), AI greetings, creative copy, and club insights call OpenAI via /api/ai/*.
- 2 UPI payment proof and expense bills are OCR'd (/api/ai/ocr) to auto-fill amounts and reconcile.
- 3 When no OpenAI key is present, every feature falls back to deterministic hand-written templates.

Automation schedule, data model & security

Vercel Cron schedule

SCHEDULE	ENDPOINT	PURPOSE
Daily 03:00	/api/cron/automation?schedule=1	Drives the automation engine — reminders, receipts, notifications.
Monthly 1st 04:00	/api/cron/reports?type=monthly	Generates the monthly activity & finance report PDF.
Quarterly 1st 05:00	/api/cron/reports?type=quarterly	Quarterly consolidated report.
Half-yearly 1st 06:00	/api/cron/reports?type=half_yearly	Half-yearly consolidated report.
Yearly Jul 1 07:00	/api/cron/reports?type=yearly	Annual report aligned to the Lions year.
Daily 08:00	/api/cron/action-items	Surfaces & reminds outstanding zone action items.
Apr 1-7 09:00	/api/cron/donor-pack	Builds annual donor appreciation packs.
Daily 10:00	/api/cron/sync-worker	Drains the sync queue against Lions REST.

Data model — 55 migrations

Postgres schema evolves through ordered migrations (0001 -> 0055). Highlights:

- Core CRM (0001), social/creative (0002), enterprise federation hierarchy + auth + audit + sync (0003).
- Payments stack: invoices (0010), refunds (0011), portal OTP (0012), recurring invoices (0013), agent commissions (0015).
- Reporting engine (0020), push subscriptions (0021), media buckets (0023), zone agenda/minutes/calendar (0026–0028).
- Lions OIDC + REST + sandbox settings (0030–0033), cron settings (0034), governance & district governor (0035–0036).
- Federation RLS (0037), three-tier dues (0040), district sync circulars (0041), advisory voting (0046), sync queue (0048).
- Public site tables (0051), blog storytelling (0052), Lions webhook & sync meta (0053), district code 3232 F1 (0054).

Security & compliance posture

- Row-Level Security on every table; service-role key is server-only and never bundled to the browser.
- Secrets read through a Zod env schema; in-app secrets (OIDC, Lions API, OpenAI, VAPID, Cron) wrapped with AES-256-GCM at rest.
- OIDC code flow protected by state + PKCE + nonce; ID tokens verified against the provider JWKS.
- Razorpay & PhonePe webhooks HMAC-verified; Lions inbound webhook signed with a shared HMAC-SHA256 secret.
- Public POST endpoints rate-limited; append-only audit_logs capture every privileged action; RBAC denials recorded.
- Security headers (HSTS, X-Frame-Options, nosniff, Referrer-Policy, Permissions-Policy) set in vercel.json.
- No scraping — all Lions data ingested via OIDC userinfo, official REST APIs, or human-uploaded CSV/Excel.

Live view

This blueprint is mirrored in-app at /admin/workflow, and live integration health is at /admin/integrations.